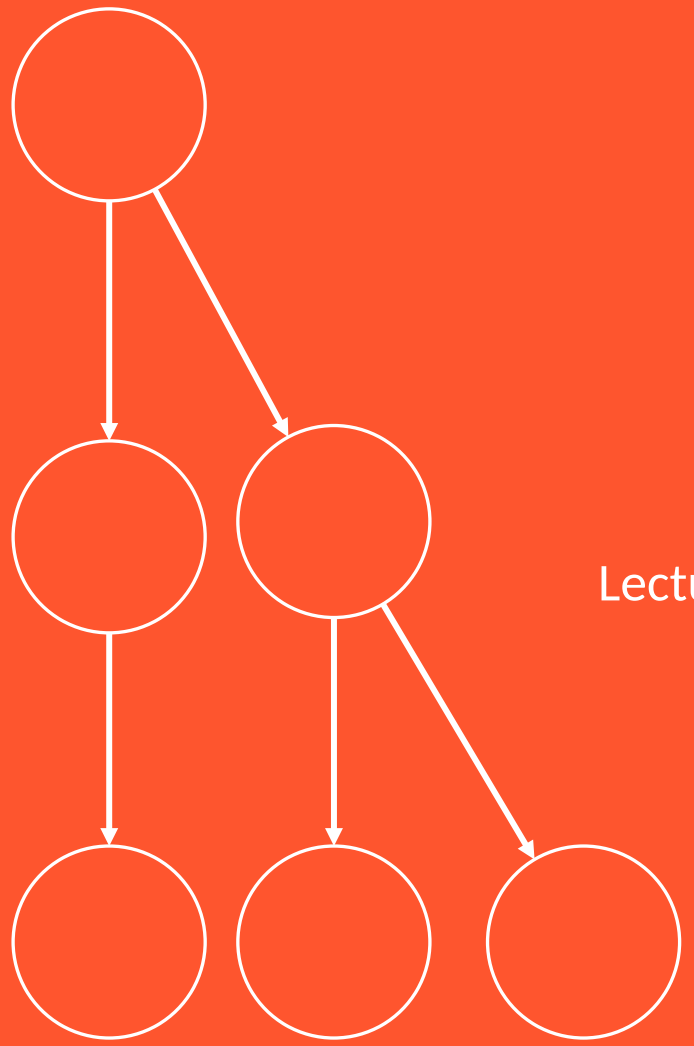


BCOG 100

Memory, Imagination, and Generative Models

Lecture 10.1 — Remembering What Happened, and Acting on What Has Not

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: How Good is Your Memory?

Which one is the real penny?



(a)



(b)



(c)



(d)



(e)

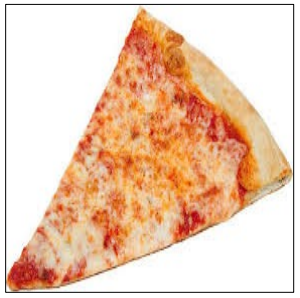


(f)

To Start: How Good is Your Memory?





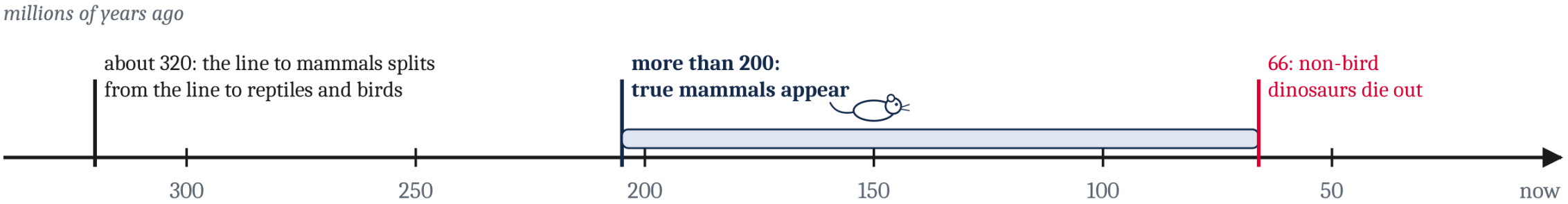







A Warm Body Needs Food, Constantly

Mammals lived alongside the dinosaurs for well over 100 million years. Their great increase in number and size came after the dinosaurs died out.

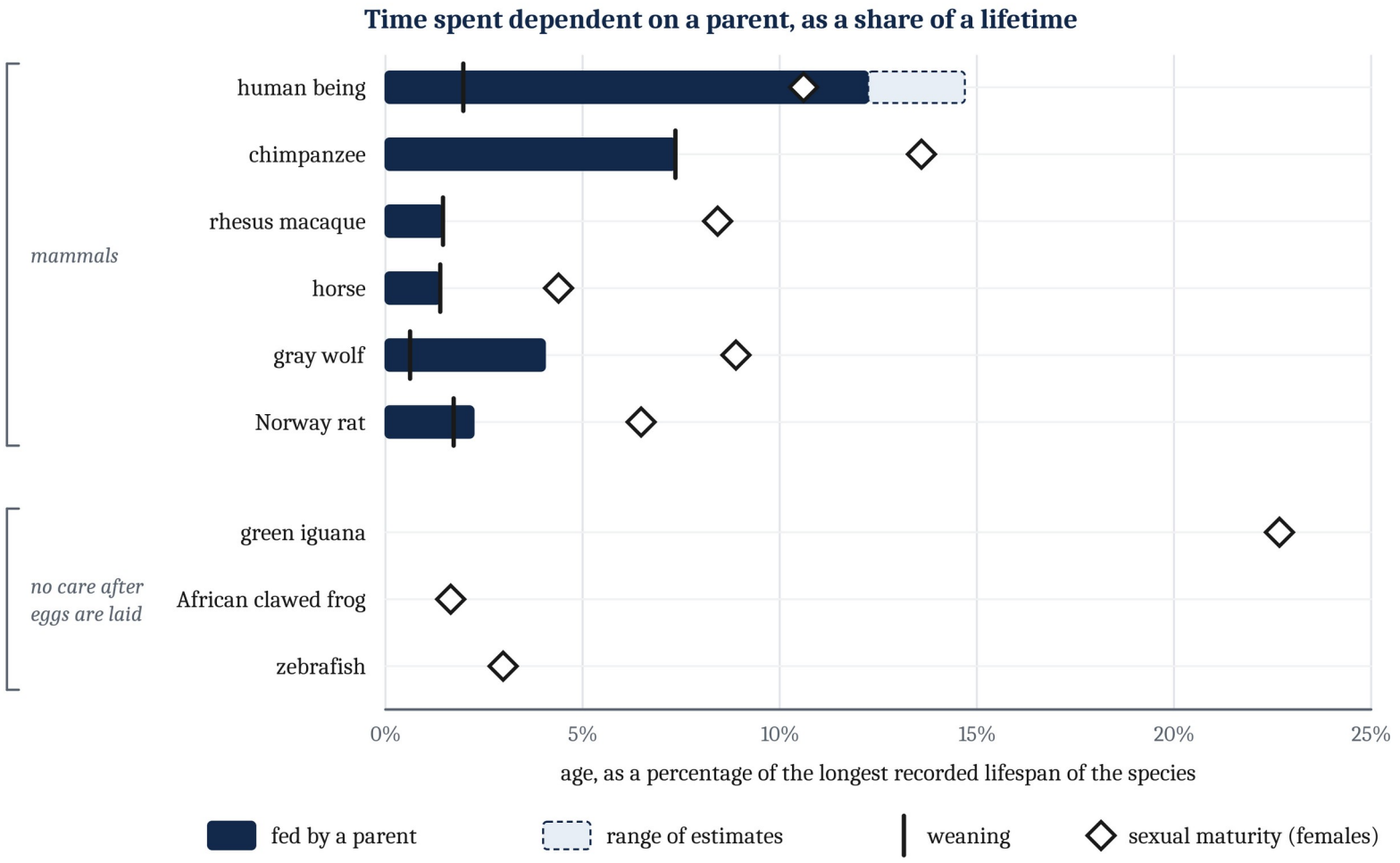


A warm body needs food, constantly

That constant need makes it useful to learn and remember facts from single occasions, and so do other features of mammals' lives: costly young, a long youth, a nest to return to, food used up where it is found, and two-state sleep.

Milk, and a Long Dependence

No newborn mammal can find enough food alone, and milk is energetically expensive. Mammals stand out for a long dependence on their mother.



What Is New in a Mammal’s Life

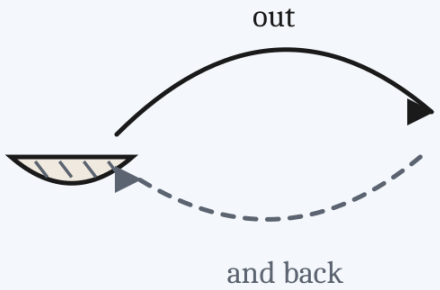
A long youth, a nest to return to, food used up where it is found, and two kinds of sleep.

Play and exploring



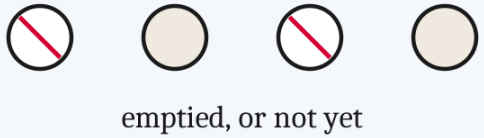
Play is part of how young animals learn and explore. Many young are more exploratory, or less wary of new things, than adults.

A nest to return to



Young stay in the nest; the mother nurses at long intervals, and returns to the same places around it.

Food used up, or hidden



Insects, seeds, fruit, and nectar are used up where found. Gray squirrels bury each nut in a separate place, and dig them up months later.

Two kinds of sleep

One state: still and quiet.

The other: head and limbs slack, eyes darting under closed lids, breathing irregular, limbs and face twitching.

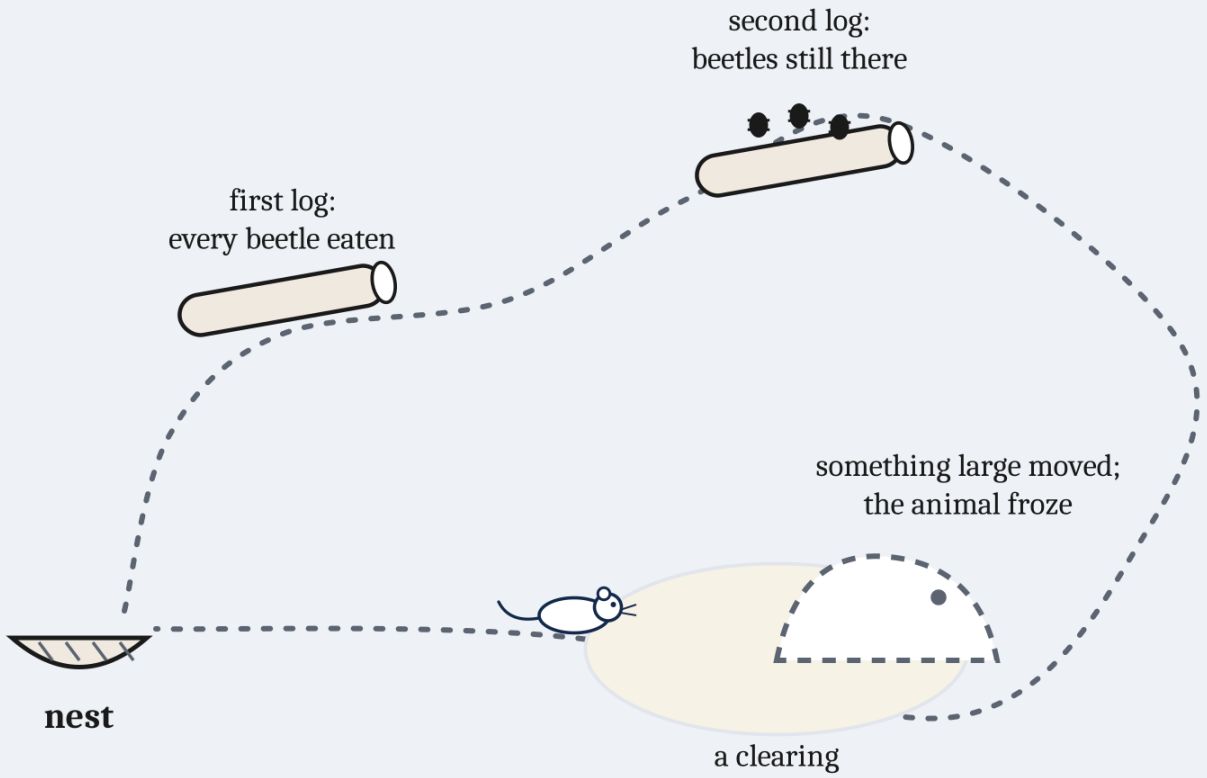
The twitching state is greatest early in life: about 8 hours a day in newborn humans, about 2 in adults.

An animal that forages over a range both constant and changing, returns to one home, and raises young through a long dependence encounters many occasions whose details are useful later.

One Night

Each fact is about one night. How to use it depends on facts that will be true the next night.

One night, as an illustration



Each fact is about one night

Not what usually happens at logs or clearings, but what happened at these ones, last night: no beetles left at the first log; some at the second; something large near the clearing.

How to use it depends on the next night

How hungry is the animal?

Did another animal find the second log first?

Is the large animal still nearby?

None of these is known on the night the facts are learned.

An illustration of a typical night for a small early mammal, as we reconstruct it.

Two Kinds of Fact

Some facts are true night after night. Others are true for one night only, and the animal learns them before it knows when they will matter.

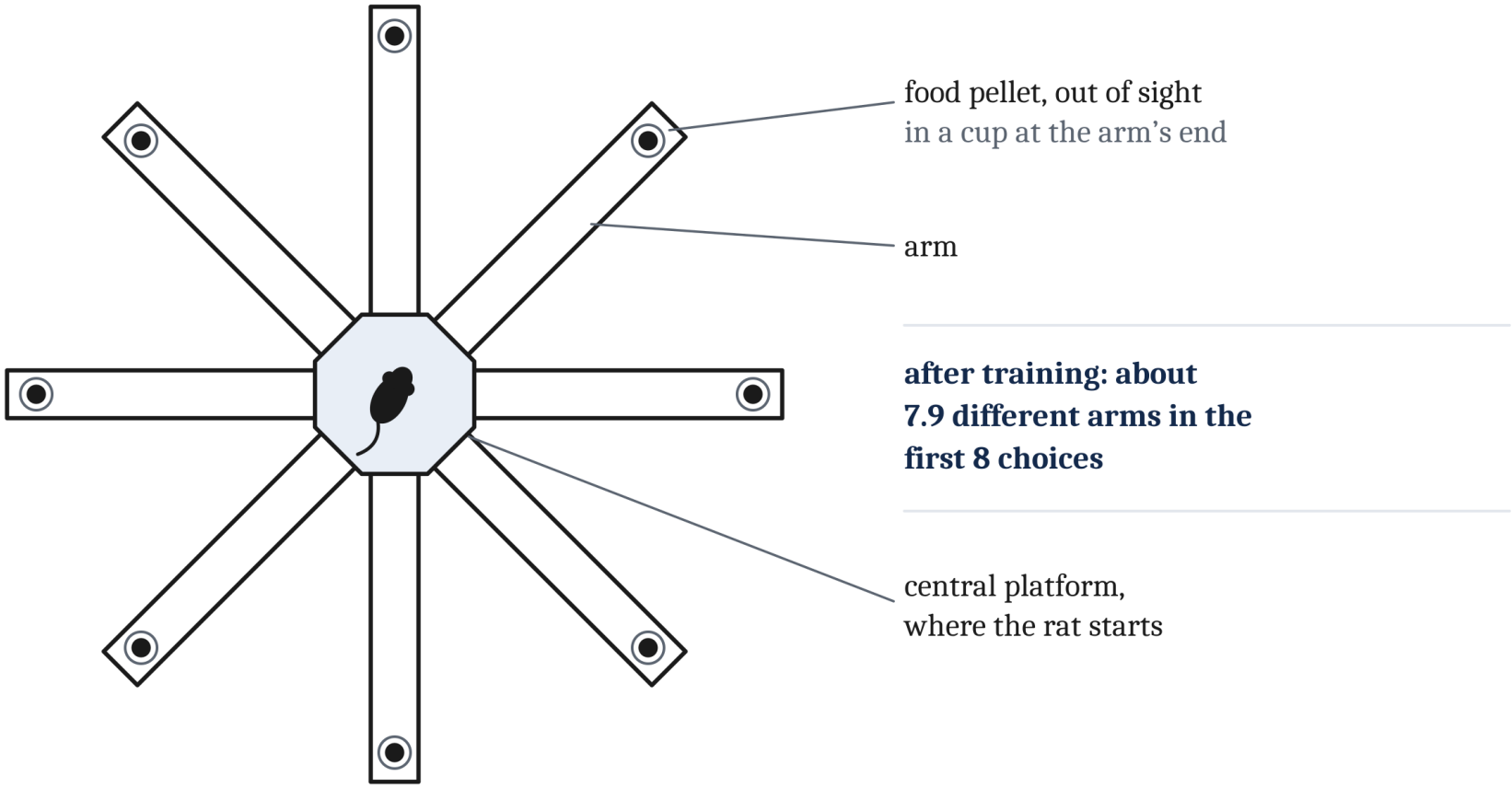
True night after night	True for one night only
Beetles are usually found under loose bark.	The beetles under this log's bark were eaten last night.
A person usually parks in one spot.	Today, the car is somewhere else.
A flower makes nectar again, slowly, after it is emptied.	This flower was emptied a little while ago.

Knowing only what is usually true does not tell an animal what happened last night, any more than knowing the usual spot finds a car parked somewhere else today. Hawaiian honeycreepers empty a flower, then stay away from it for several hours.

The Radial Maze

After training, rats entered about 7.9 different arms in their first 8 choices, and not by going around in a fixed order.

An eight-arm radial maze



from above; after Olton (1979), Fig. 5 and p. 588

For a Moment, for a Task, for a Lifetime

Three problems differ in how long, and for what, an animal must be able to use what it learned.

How long it must be used	*What is used*	*The early mammal*
For a moment	what was just sensed	*the rustle it just heard*
For seconds to minutes	what a task needs, while it is done	*which side of this log it has searched*
For days, months, or years	what was learned earlier	*the way back to its nest*

The first two may need only a temporary state that ends with the task.
The third requires a change that outlasts the experience.

What Remembering Requires

Five things any system that learns particular occasions must do, whether it is an animal or a machine.

1

Recall a whole from a part

the smell of one log is enough to recall that night there

2

Tell similar occasions apart

two nights at the same log

3

Learn new, recall old

learn about new nights and still recall old ones

4

Recall across a lifetime

some occasions, for the rest of its life

5

Use it in a way not foreseen

in a situation that could not have been predicted

Required of any such system, an animal or a machine

An animal: well enough

It remembers well enough if it chooses well the next night, even without recalling every detail of the night before.

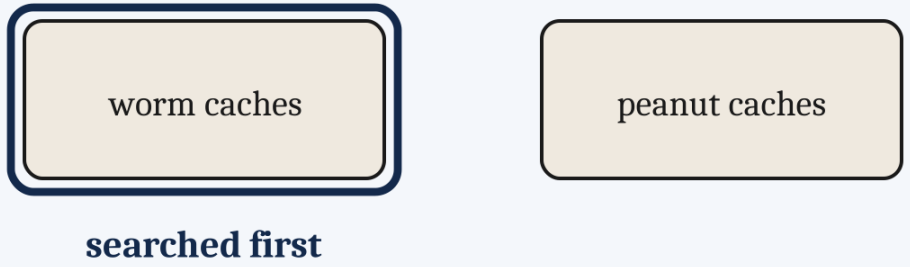
A computer: an exact copy

It is used where an exact copy is required, as with a bank balance or a document.

Testing for One Occasion

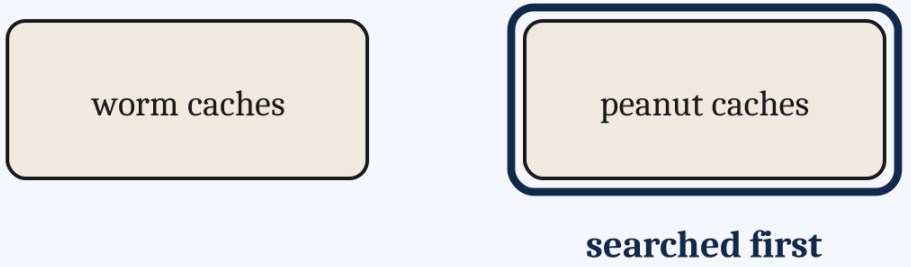
A test the animal could not have anticipated is hard to explain as a prepared response. After four hours, scrub jays searched first for worms; after five days, for peanuts.

Worms cached 4 hours earlier



the worms are still good

Worms cached 5 days earlier



the worms would have rotted

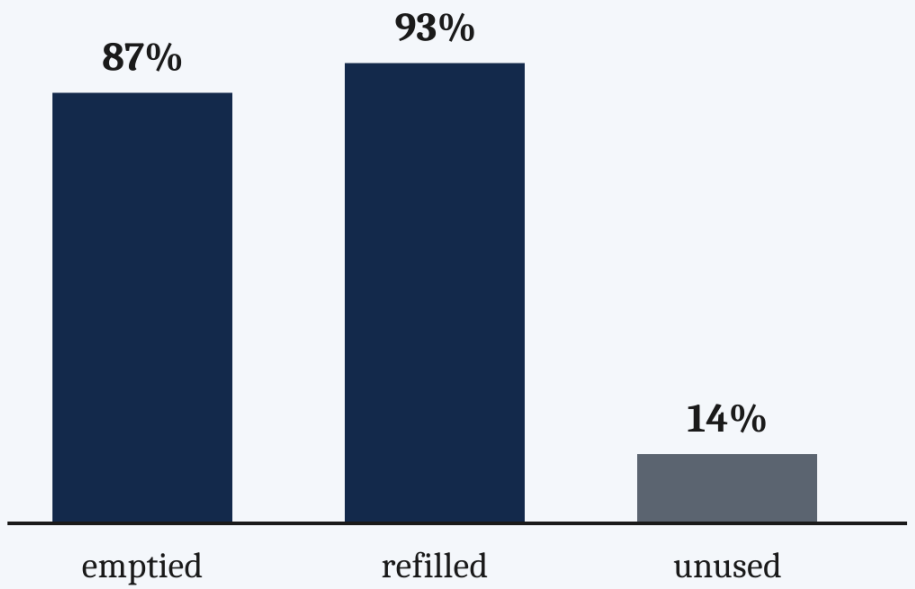
Behavior alone may not separate information about one occasion from a weaker tendency. A test the animal could not have anticipated is hard to explain as a prepared response, but not impossible, so researchers add controls and look for several kinds of evidence.

What Animals Do After One Occasion

In each study, behavior depended on one particular occasion after it was over.

Kangaroo rats, a day after caching

Share of cups searched; odors removed



Emptied sites searched almost as often as refilled ones, so the smell of the seeds cannot explain it.

Other tests after one occasion

Pigeons

chose the color matching what they had just done on about 72% of first tests (chance: 50%)

Dogs

unexpectedly told to imitate: 11 of 17 did after one minute, 6 of 17 after an hour

Rats

a fork appeared without warning after foraging; they turned the way that matched whether they had found food

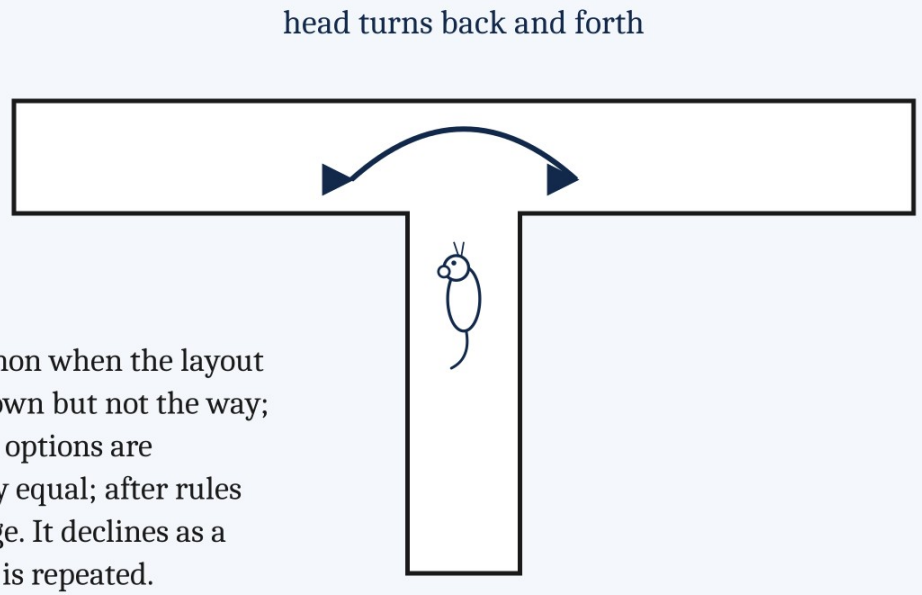
Rats, monkeys

picked out odors by their order in one list; learned from one trial in a scene of its own

Acting on What Has Not Happened

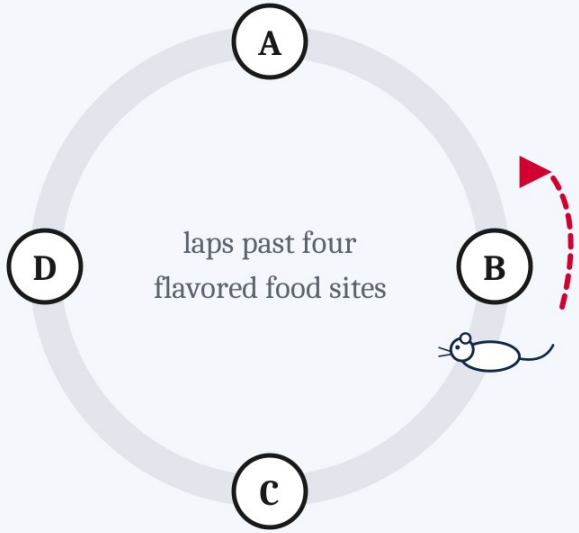
Vicarious trial and error: a rat pausing at a fork and turning its head back and forth between the paths before it chooses.

Pausing at a fork



Common when the layout is known but not the way; when options are nearly equal; after rules change. It declines as a route is repeated.

Looking back at a skipped site



Skipped a site it would usually wait at, then reached a longer delay: it looked back, then waited through the long delay more often than usual.

To Discuss: Did the Dog Remember?

In your group, five minutes.

A friend says her dog remembers where she hid a treat yesterday.

The dog might be using that one occasion. Or it might be following a habit, or simply the smell of the treat.

Design a test that would tell these apart. What would the dog have to do, on a test it could not have anticipated? What control would you add?

Image Sources (1 of 2)

Slide	Image	File	License	Source
3	One night, and what its use depends on	fig-one-night.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
4	Facts true night after night, and for one night only	fig-two-kinds.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
5	An eight-arm radial maze	fig-radial-maze.png	CC BY 4.0	Jon Willits — from the Module 10 reading, after Olton (1979) · own work
6	When mammals arose and spread	fig-mammal-timeline.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	Time spent dependent on a parent, as a share of a lifetime	fig-dependence.png	CC BY 4.0	Jon Willits — from the Module 10 reading; data from the AnAge database and published field studies · own work
8	Four features of a mammal's life	fig-mammal-life.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
9	Three time scales	fig-time-scales.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
10	Five requirements, and how well is well enough	fig-requirements.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

Image Sources (2 of 2)

Slide	Image	File	License	Source
11	Scrub jays' worms and peanuts	fig-jays.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
12	Caches, and tests of a kind never trained	fig-one-occasion-tests.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	Pausing at a fork, and looking back	fig-vte.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work